

Name :

ID :

Groupe:

Exercice 1. Among these assertions, specify which ones are true and which ones are false.

- 1- The sample mean is always higher than the population mean.
- 2- The median is always higher than the mean.
- 3- The mean of (1, 2, 8, 9) is 10.
- 4- The STD of a sample is always less than 1.
- 5- The STD of a sample can be -1.
- 6- When median > mean then the distribution is symmetrical.
- 7- Increasing all values of a sample by 5, will increase the mean by 5.
- 8- Increasing all values of a sample by 5, will increase STD by 5.
- 9- Multiplying all values by 2, will multiply the mean by 2.
- 10- Multiplying all values by 2, will multiply the STD by 2.
- 11- The mean is at the middle of the histogram.
- 12- There is 50% chance that one random sample will be in the IQR. ($IQR = Q_3 - Q_1$)
- 13- We can learn the IQR by looking at the boxplot.
- 14- If a value is not in the IQR, then it is an outlier.
- 15- If a value is higher than the mean then it is an outlier.
- 16- Removing an outlier that is positive, will increase the mean.
- 17- Removing an outlier that is negative, will most likely decrease the STD.
- 18- A histogram of any distribution is bell-shaped.
- 19- On the x-axis, a histogram has always positive values.
- 20- The mean is equal to 0 for a symmetric histogram.

0.25pts x 20 = 5 pts

True	False
	✓
	✓
	✓
	✓
	✓
	✓
✓	
	✓
✓	
✓	
	✓
✓	
	✓
	✓
	✓
✓	
	✓
	✓
	✓

Exercice 2. We took note of the performance bonuses (in DA) of the employees of a company. The results are given in the following statistical table :

Bonuses	frequency	Rel. Frequency	Cum. Frequency	Cum. Rel. Frequency
[750, 1250[	26	0.130	26	0.130
[1250, 1750[	71	0.355	97	0.485
[1750, 2250[	43	0.215	140	0.700
[2250, 2750[	26	0.130	166	0.830
[2750, 3250[	18	0.09	184	0.920
[3250, 3750[	10	0.05	194	0.970
[3750, 4250[	6	0.03	200	1.00

1. Complete the statistical table above.
2. What is the bonus value received by 50% of the employees.

Bonus value received by 50% of the employees ($n = 200$) represents the median $\tilde{X} \in [1750, 2250[$ and given by

$$\tilde{X} = L_1 + \left(\frac{\frac{n}{2} - C f_b}{f_m} \right) \times w = 1750 + \left(\frac{\frac{200}{2} - 97}{43} \right) \times (2250 - 1750) = \mathbf{1784.9}$$

3. Give Q_1, Q_3 and interquartile range

The first quartile Q_1 is located in the class $[1250, 1750[$ and the third quartile Q_3 is in $[2250, 2750[$ and both are given by

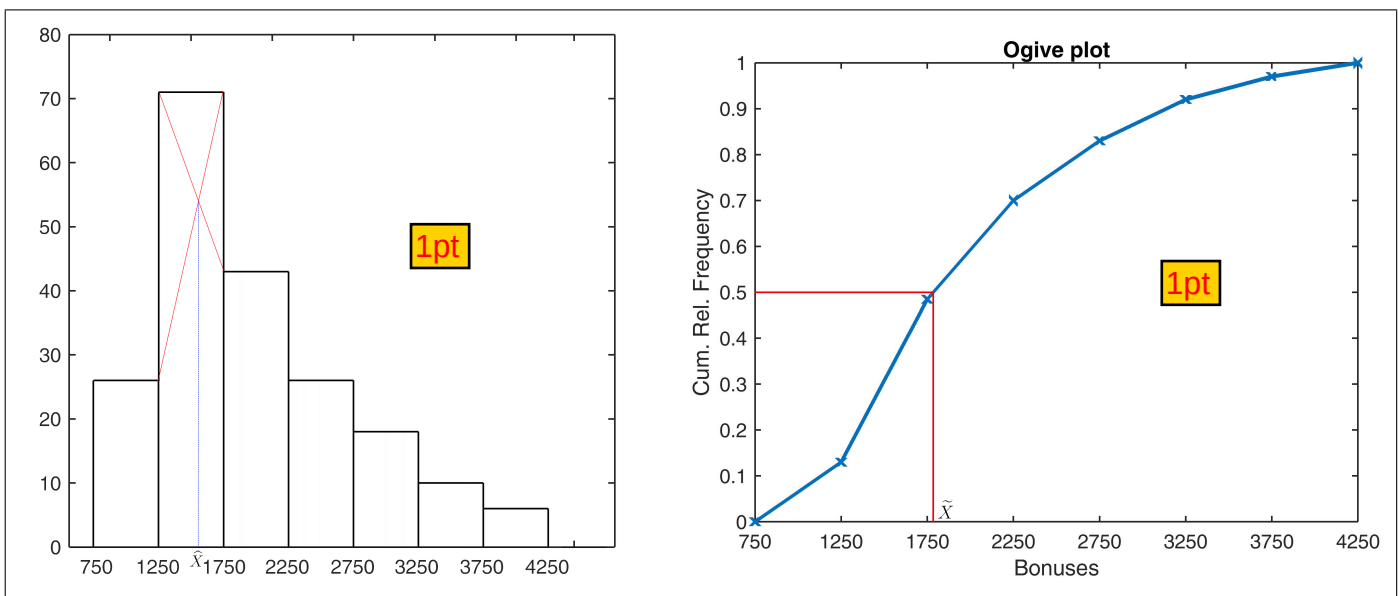
$$Q_1 = L_1 + \left(\frac{\frac{n}{4} - C f_{q_1}}{f_{q_1}} \right) \times w = 1250 + \left(\frac{\frac{200}{4} - 26}{71} \right) \times (1750 - 1250) = \mathbf{1419} \quad \mathbf{0.5pts}$$

and

$$Q_3 = L_1 + \left(\frac{\frac{3n}{4} - C f_{q_3}}{f_{q_3}} \right) \times w = 2250 + \left(\frac{\frac{600}{4} - 140}{26} \right) \times (2750 - 2250) = \mathbf{2442.3}. \quad \mathbf{0.5pts}$$

The interquartile range is $IQR = Q_3 - Q_1 = 2442.3 - 1419 = \mathbf{1023.3}. \quad \mathbf{0.25pts}$

4. Plot the histogram and the Ogive plot of the above distribution then determine the mode and median of the series graphically



5. Give the modale class then calculate the mode value.

Clearly, The highest frequency is for the second class $[1250, 1750[$ to which the mode $\hat{X}$ belongs and it's given by

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 1250 + \left(\frac{71 - 26}{(71 - 26) + (71 - 43)} \right) \times (1750 - 1250) = \mathbf{1558.2} \quad \mathbf{0.5pts}$$

6. Calculate the mean, variance, standard deviation and coefficient of variation.

Since the data are grouped, we need to compute the med point of each class and the formulas to be used are :

$$\mu = \bar{X} = \frac{\sum_{i=1}^7 f_i x_i}{\sum_{i=1}^7 f_i} = \frac{396500}{200} = \mathbf{1982.5} \quad \mathbf{0.5pts}$$

$$\sigma^2 = Var(X) = \frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2 = \frac{8874643.75}{200} = \mathbf{44373.22} \quad \mathbf{1pt}$$

and

$$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{44373.21875} = \mathbf{210.65}. \quad \mathbf{0.25pts}$$

The coefficient of variation is given by

$$CV = \frac{\sigma}{\mu} \times 100 = CV = \frac{210.65}{1982.5} \times 100 = \mathbf{10.625\%}$$

0.5pt

7. Do you think the distribution of the performance bonuses is symmetric? justify.

No, the distribution of the performance bonuses is not symmetric characterized by the shape of the histogram, also by computing pearson's coefficient of skewness

$$sk = \frac{3 \times (1982.5 - 1784.9)}{210.65} = \mathbf{2.8141}$$

0.5pts

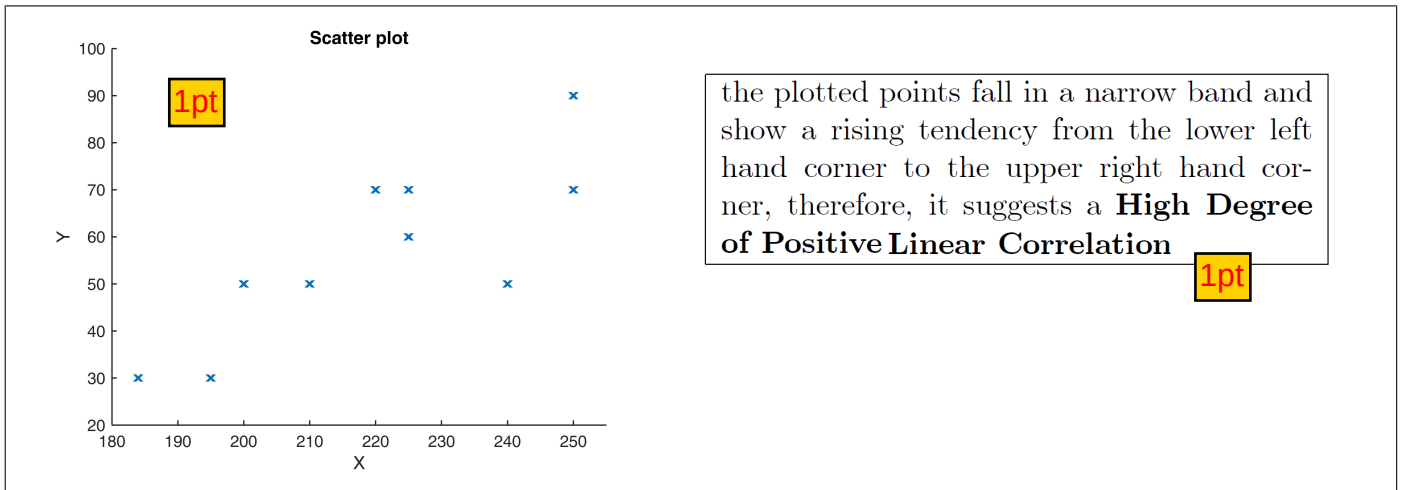
we can observe that its value is close to 3 which let us conclude that the distribution of the performance bonuses is asymmetric

Exercise 3.Wheat production (X in quintals) and the number of rainy days (Y) were recorded. The following results were obtained:

X	200	184	225	250	240	195	210	225	250	220
Y	50	30	70	90	50	30	50	60	70	70
$(x_i - \bar{X})^2$	396.01	1288.81	26.01	906.01	404.01	620.01	98.01	26.01	906.01	0.01
$(y_i - \bar{Y})^2$	49	729	169	1089	49	729	49	9	169	169
$(x_i - \bar{X})(y_i - \bar{Y})$	139.3	969.3	66.3	993.3	-140.7	672.3	69.3	15.3	391.3	1.3

$$\sum x_i = 2199, \sum y_i = 570, \sum (x_i - \bar{X})^2 = 4670.9, \sum (y_i - \bar{Y})^2 = 3210, \sum (x_i - \bar{X})(y_i - \bar{Y}) = 3177.$$

1. Draw the scatter plot of the given data. Describe the relationship between X and Y .



2. Calculate the correlation coefficient and interpret the obtained result.

The above scatter plot suggested the presence of linear correlation, hence, we choose pearson's coefficient of correlation to measure it.

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2} \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{3177}{\sqrt{4670.9} \times \sqrt{3210}} = \mathbf{0.82047}$$

3pts

which indicates the existence of a strong positive linear correlation between the two variables.

Exercice 4. Let X be a numeric and discrete statistical variable with the modalities $x_1, \dots, x_n$. Denote by $f_1, \dots, f_n$ the associated frequencies.

Show that :

$$\sum_{i=1}^n f_i(x_i - \bar{X})^2 = \min_{a \in \mathbb{R}} \sum_{i=1}^n f_i(x_i - a)^2$$

Set

$$F(a) = \sum_{i=1}^n f_i(x_i - a)^2$$

clearly F is differentiable, hence, we calculate F' then we solve the equation $F'(a) = 0$ for a possible minimum(s) or maximum(s) by checking the signe of F'' .

$$\begin{aligned} F'(a) &= - \sum_{i=1}^n f_i(x_i - a) = 0 \implies - \sum_{i=1}^n f_i x_i + \sum_{i=1}^n f_i a = 0 \\ \implies a \sum_{i=1}^n f_i &= \sum_{i=1}^n f_i x_i \implies a = \frac{1}{\sum_{i=1}^n f_i} \sum_{i=1}^n f_i x_i = \bar{X}, \end{aligned}$$

moreover, $\forall a \in \mathbb{R}, F''(a) = \sum_{i=1}^n f_i = 1 > 0$ which means the F has only one minimum at $a = \bar{X}$, thus

$$\sum_{i=1}^n f_i(x_i - \bar{X})^2 = \min_{a \in \mathbb{R}} \sum_{i=1}^n f_i(x_i - a)^2$$

2pts